
Forecasting Chaos Across Model Families: An Out-of-Sample, Reliability-Adjusted Benchmark

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Abstract

This paper evaluates chaotic forecasting across model families under a shared out-of-sample protocol. The benchmark is a controlled k -pendulum system ($k \in \{1, 2, 3\}$; chaotic for $k \geq 2$) with four families: large language models (LLMs), Chronos, learned dynamics models (Neural ODE, HNN, LNN), and classical integrators. Reliability-adjusted scoring assigns a fixed error to refusals and parse failures rather than discarding them. Off their training trajectories, learned dynamics models increase from ~ 0.26 to 0.97 – 1.14 rad for the stable variants and do not outperform the linearized-dynamics or nearest-neighbor baselines. Among the evaluated black-box methods, *kimi* has the lowest error (0.538 rad), while known-physics integrators remain the numerical reference. Objective effects are architecture-dependent: mixed- k rollout is lower for the Neural ODE, whereas single- k rollout is lower for the HNN. A matched disclosure experiment and inference probe find little evidence that the LLMs infer hidden physical constants; under this protocol they default to Earth-standard values. Cost is also asymmetric: the LLM run is the only per-query billed evaluation. The results indicate that in-sample evaluation can overestimate learned-dynamics performance and motivate controlled out-of-sample comparison of scientific forecasting methods.

1 Introduction

Comparing forecasting methods on a chaotic system depends on evaluation design: whether models are scored in or out of sample, how failures are handled, and what information each model receives. We evaluate LLMs, time-series foundation models, learned dynamics models, and classical integrators on a shared k -link pendulum benchmark. Given state s_t , the task is to predict s_{t+T} . The pendulum is integrable at $k=1$ and chaotic for $k \geq 2$, has high-precision ground truth, and requires no domain-specific grading.

Contributions.

1. A **cross-family evaluation protocol** in which all families are scored on the same out-of-sample held-out trajectories, with a reliability-adjusted metric that assigns a fixed error to abstentions rather than discarding them (§3.4, §5.1).
2. An **out-of-sample leaderboard**: tested off their training trajectories the learned models reach 0.97 – 1.14 rad for the stable variants (from ~ 0.26 in-sample) and do not outperform the linearized-dynamics or nearest-neighbor baselines, while among the evaluated black-box methods the lowest error is obtained by *kimi* (0.538 rad). Objective effects are architecture-dependent rather than uniformly favorable to one training loss (§5.4).

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3. **Chaos-appropriate analysis:** a predictability-horizon metric and a confound-free system-identification test. An unadjusted regime contrast suggests lower error when constants are hidden; a matched paired experiment and a direct inference probe indicate that this is a difficulty confound rather than parameter inference (§5.6).

2 Related Work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias through a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler–Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds [Gilpin, 2021]. We evaluate all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialized as text [Gruver et al., 2023], and time-series foundation models—Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024]—are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. We evaluate these forecasters on the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs perform well on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]; recent benchmarks probe physics reasoning in LLMs [Xu et al., 2025]. Our task instead requires quantitative trajectory prediction of a chaotic system rather than qualitative or symbolic reasoning.

3 Benchmark Design

Choice of system. We use the k -link pendulum to study evaluation methodology, not pendulums as an application domain. It provides a tunable chaos axis, high-precision ground truth, known equations for numerical references, and controllable constants for the system-identification test (§5.6). The single controlled system is therefore a deliberate design choice.

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to predict $\mathbf{s}_T$ at horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4 at this step, the RK4 baseline reported later (the same integrator at a $10\times$ coarser $\Delta t=10^{-2}$) is a numerical reference by construction rather than an independent competitor; Euler and the symplectic integrator are distinct known-physics methods. Inputs available to each family are stated in §3.3.

3.2 Regimes and grid

Three regimes test generalization to non-standard constants (Table 1). In `changed_hidden` the constants (g, L, m , damping) are not disclosed to the model, which makes the cell an in-context system-identification problem. An evaluation cell is a tuple (model, k , regime, modality, horizon, prompting); horizons are $\{0.01, 1, 10, 60\}$ s, modalities $\{\text{coords}, \text{image}, \text{image+coords}\}$ (LLMs only), and prompting $\{\text{direct}, \text{chain-of-thought}\}$ (LLMs only). Non-LLM models use the coordinate/direct setting.

Out-of-sample protocol. Learned dynamics models are evaluated off their training data. We generate a shared held-out set (a separate seed from the training trajectories) of 180 initial conditions (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate every family on the identical trajectories, keyed by `movement_id`. The cross-family leaderboard (§5.1) uses horizons $\{1, 10\}$ s, where chaos has

Table 1: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
<code>changed_disclosed</code>	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
<code>changed_hidden</code>	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 2: Inputs available to each family. State is at $t=0$; history is $\text{pre-}t=0$. Chronos is history-conditioned; learned hidden cells withhold hidden constants.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except hidden	—

begun but error has not saturated (§3.4), giving $180 \times 2 = 360$ cells per model. We report no in-sample numbers and quantify the size of the in-sample effect directly (§5.1).

3.3 Inputs and grouping

Table 2 states what each family receives. Chronos is a time-series model, so we evaluate it in its standard form: a history-conditioned forecaster with a 10 s $\text{pre-}t=0$ context window. The LLMs and learned dynamics models are state-conditioned from $t=0$. For learned models, hidden-regime cells are equalized by withholding hidden constants and using normal-regime prior constants. A separate asymmetry runs in the opposite direction: the LLMs are large-scale proprietary models, whereas the learned models are small (256-wide, 3-layer MLPs trained on a single 8 GB GPU; Appendix B). The behavior of larger physics-informed networks is outside the scope of this study. We separate known-physics methods (RK4, Euler, symplectic) from black-box methods (LLMs, Chronos, learned dynamics); the former are numerical references.

3.4 Metrics

The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories; we also track Cartesian bob error, angular-velocity sign match, and energy drift. Three choices are specific to this setting.

Reliability-adjusted error. A cell is answered only if the model returns a parseable, finite prediction. Reporting error over answered cells alone would reward abstention on hard cells. We therefore assign unanswered cells the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform guess); we also report the strict upper bound π . Answering is thus part of the score.

Predictability horizon. For chaotic systems, pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that exceeds τ . The contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. We sweep τ .

System-identification gap. We define $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$. Positive values indicate a penalty for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty, we also run a matched experiment (§5.6) that toggles disclosure on identical trajectories and asks hidden-condition models to report inferred constants.

4 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (Azure AI Foundry deployments; evaluated June 2026), with a structured prompt (system description, disclosed constants where applicable, initial state) and a JSON answer schema (Appendix A). Each cell uses one temperature-0 call with a uniform token and timeout budget; no best-of- N aggregation is used. The LLM evaluation (Parts A and B, 3,960 cells) records prompt and completion token counts and has an estimated nominal API cost of \$110.38 under the pricing assumptions in the released analysis script.

Time-series foundation models. We run Chronos [Ansari et al., 2024] locally on a single GPU in its standard history-conditioned form: the 10 s pre- $t=0$ context window of all $2k$ state channels, resampled to up to 512 points, forecasts to the target horizon (prediction length ≤ 64 steps). The leaderboard uses `chronos-2` in univariate and multivariate modes; the latter forecasts all $2k$ channels jointly. The paired mode comparison uses 20 trajectories per (k, regime) cell. The earlier `chronos-t5-large` was preliminary; Moirai’s `uni2ts` could not be installed in our environment.

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`. We compare two objectives: derivative matching (MSE on finite-difference $\dot{s}$) and rollout loss (MSE on the state after k Euler steps, with gradients through the unroll). We also use mixed- k rollout, which trains on several window lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order autodifferentiation, which Apple’s MPS backend does not run; we train it at full scale (67k samples, hidden 256) on a CUDA GPU. All learned models are trained on the same nine trajectories—one per (k, regime) cell (Appendix B), a small budget for a chaotic system.

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is from discretization alone. RK4 is the data-generating process and serves as a numerical reference rather than an independent competitor.

Baselines. Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast from the matching (k, regime) training trajectory. The nearest-neighbor baseline uses the same data as the learned models.

5 Results

5.1 Out-of-sample leaderboard

Table 3 reports the out-of-sample leaderboard on the shared held-out set (360 cells: 180 trajectories $\times$ horizons $\{1, 10\}$ s), using the reliability-adjusted metric. Every model is scored on the same grid; the Answered column is the fraction of valid predictions. LLMs use direct-prompt forecasts. Confidence intervals are 95% paired bootstrap over trajectories ($B=10^4$); for LLMs they do not include per-call sampling variation.

The classical integrators have the lowest error: RK4 (the data-generating process) at 0.059, symplectic at 0.205, Euler at 0.405. Among the black-box methods evaluated, the lowest error is obtained by `kimi` (0.538 rad, 99% answered), below both Chronos modes and all learned dynamics models. The four baselines provide reference scores. `kimi` is below every baseline with a non-overlapping confidence interval (the closest, nearest-neighbor, is 0.725), so its standing is not explained by the scoring rule; both Chronos modes are also below the nearest-neighbor baseline in point estimate. The learned dynamics models are not: after replacing the hidden-regime cells with the equalized no-hidden-constants run, the nearest-neighbor analog lookup (0.725) and the linearized-pendulum baseline (0.855) are below every learned model. Persistence (1.004) is also below all learned models except HNN-rollout (0.972), with intervals overlapping for the closest comparisons. Out of sample the learned models therefore do not outperform the linearized or nearest-neighbor baselines, despite those baselines using the same training data. They score ~ 0.26 rad under in-sample evaluation and reach 0.97–1.14 rad here for the stable learned models; this factor of approximately four indicates

Table 3: Out-of-sample leaderboard on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at $\pi/2$; 95% bootstrap CI. RK4 is the data-generating process and is reported as a numerical reference. Methods new in this work are marked $\dagger$; simple baselines are marked $\ddagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor $\ddagger$	0.725	[0.657, 0.797]	100%	baseline
7	linearized $\ddagger$	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence $\ddagger$	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed $\dagger$	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed $\dagger$	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity $\ddagger$	1.319	[1.241, 1.396]	100%	baseline
18	lnn $\dagger$ (full-scale)	1.349	[1.283, 1.416]	66%	learned

that their in-sample strength on this benchmark is largely attributable to evaluation on their training trajectories.

Effect of the reliability adjustment. grok answers 82% of the grid and lnn 66%; scoring those abstentions at $\pi/2$, rather than averaging over answered cells, places them below more reliable methods. Averaging over answered cells alone would rank the least reliable models higher.

5.2 Behavior by horizon

Out-of-sample error by horizon appears in Appendix D (Table 9). At 1 s the ordering follows the leaderboard: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.86–1.26. At 10 s all models are in the chaotic regime, and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.44. The mixed- k Neural ODE, which is the in-sample standout at 1–10 s, is mid-pack here (0.951 / 1.120). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the leaderboard is the range over which the families remain separable.

5.3 Breakdown by k and by regime

Tables 7 and 8 (Appendix C) report out-of-sample error by number of links and by regime. Error increases monotonically with k for every family (for kimi, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The changed_hidden regime has the lowest error for almost every model (for kimi, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the hidden constants reduces but does not remove the same difficulty effect: hidden-regime errors are now 0.71–1.19 rather than the previous, true-constant-favored values. The low-gravity, short-rod constants produce slower, less-divergent motion over a fixed horizon, so the unadjusted lower hidden-regime error contrast is a difficulty confound (§5.6). Re-estimating the leaderboard confidence intervals with a cluster bootstrap over whole held-out trajectories changes no interval by more than ~ 0.03 rad and changes no ranking, so the gaps are not a result of treating correlated cells as independent.

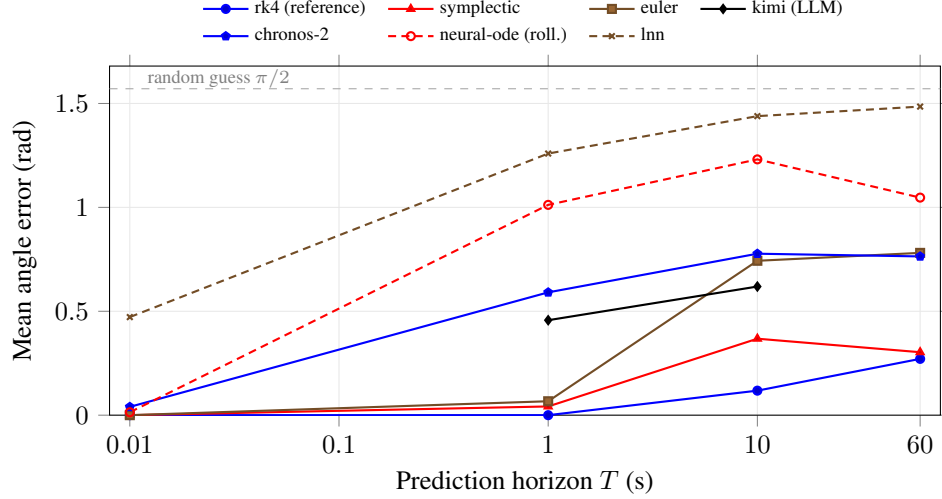


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 4: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed- k	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed- k	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

5.4 Training objective

Table 4 reports the effect of the loss with architecture fixed. Under the equalized hidden-constant analysis, rollout loss is not uniformly better: mixed- k rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.349), with 34% of cells missing or diverging under autograd-ODE integration. We make no architecture-wide claim about LNNs; under our implementation and protocol the LNN was numerically unstable and generalized poorly out of sample.

5.5 Physical fidelity: energy conservation and stability

Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (normal) regime, where true energy is constant. Table 5 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded rollouts. The learned models do not match classical integrators on this axis: the lowest-drift learned model is 0.78, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged, indicating error in the learned Hamiltonian rather than in the rollout integrator. At matched objective, HNN drift is also comparable to Neural ODE drift; the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the leaderboard.

Table 5: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^*=g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. Rolling the HNN out with a symplectic integrator gives near-identical drift, so the error is in the learned Hamiltonian, not the integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 6: Matched, confound-free system-identification gap (rad), paired on identical trajectories. $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected.

Model	disclosed	hidden	Δ_{sysid}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

5.6 Chaos-appropriate analysis

Predictability horizon. Appendix D (Table 10) reports the longest horizon each model tracks contiguously below threshold τ . Only numerical integrators stay below strict thresholds past 1 s. Every learned model returns to the first rung (0.01 s) at strict thresholds and reaches 60 s only at $\tau=1.0$ ($\sim 57^\circ$). The LLM ladder is coarser ($\{1, 10\}$ s); within it, only kimi is below 0.5 rad at 1 s.

Unadjusted regime contrast. On the shared held-out set, regime contrasts confound disclosure with trajectory difficulty. Pooled across LLMs, the hidden regime has lower error than the disclosed regime (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524 , $p_{\text{Holm}} < 10^{-3}$), reproducing the earlier “hidden is easier” pattern.

Matched experiment. The matched test toggles disclosure on identical trajectories (Table 6). The effect reverses: hiding constants increases error ($\Delta_{\text{sysid}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$). Disclosure helps at $k=1$ and at the 1 s horizon, while $k=3$ shows a small reverse effect. The unadjusted hidden-regime advantage is therefore a difficulty confound.

Inference probe. In the hidden condition, models also report inferred constants. They do not recover them: inferred- g MAE is 1.81 (deepseek), 1.88 (grok), and 2.12 (kimi) for true $g \in [1.6, 12]$. Outputs mostly default to Earth-standard values (deepseek: $g \approx 9.81$; grok/kimi: $L=m=1.0$). The probe therefore gives little evidence that regime accuracy differences are caused by in-context system identification.

5.7 Univariate versus multivariate time-series modeling

The leaderboard uses one trajectory per cell, which resolves the large gaps between families but not small effects. To test whether a time-series model that shares information across the $2k$ state channels benefits from their coupling, we compare Chronos-2 in univariate and multivariate modes with 20

trajectories per (k, regime) cell ($n=720$ paired samples) on identical trajectories. The two modes are not distinguishable: mean error 0.549 (uni, 95% CI [0.505, 0.593]) vs. 0.543 (multi, [0.498, 0.589]). The paired difference is +0.005 rad overall, with no significant effect at any k or at horizons 1 s and 10 s; the per- k trend at $n=1$ (multivariate appearing to help most at $k=3$) goes to zero at $n=720$. A nominal overall $p=0.025$ is driven by a sub-0.01 rad bias at the 0.01 s horizon and is not significant after Holm correction. We report a null result: cross-channel coupling gives no measurable benefit for time-series forecasting of this system.

6 Discussion

In-sample evaluation. Scored on held-out rather than training trajectories, the learned dynamics models increase their error and rank below a zero-shot time-series model, a zero-shot LLM, and simple baselines. The in-sample ranking is not supported under held-out evaluation; a learned simulator must be reported on trajectories it never saw. After equalizing hidden-regime constants, objective effects are model-dependent rather than uniformly explained by a single training loss.

Reliability. Scoring abstentions at $\pi/2$, rather than averaging over answered cells, lowers grok and the LNN; chain-of-thought has no significant paired effect.

System identification. The apparent lower error when constants are hidden is a difficulty confound. On matched trajectories, hiding constants slightly increases error, and the probe shows default Earth-standard values. Under this protocol, we find little evidence that the evaluated models infer hidden physical parameters.

Cost and scope. The LLM run is the only per-query billed evaluation: the released per-cell records contain prompt and completion token counts, giving an estimated nominal API cost of \$110.38 for 3,960 cells ($\sim \$0.03/\text{forecast}$ under the June 2026 deployments, prompts, and pricing assumptions). Numerical, learned, and Chronos runs have no per-query API charge when run locally, but still use local compute. Rankings should be expected to depend on the system; the transferable part is the evaluation protocol: out-of-sample scoring, reliability adjustment, and matched disclosure control.

7 Limitations

All results use one held-out seed; bootstrap intervals quantify trajectory variation, not variation across independently generated benchmark seeds. Paired tests are Holm-corrected within each family (objective, multivariate, system-ID) but not across the paper, so marginal results should be read as suggestive. Predictions are single-endpoint; full-rollout, probabilistic, attractor-distribution, and noisy-initial-condition scoring are left to future work. The comparison is structurally asymmetric: learned models are trained on the k -pendulum family, Chronos is history-conditioned, and API LLMs are much larger models with undisclosed pretraining budgets. The learned models also use only nine training trajectories, so data- and capacity-scaling remain open. HNNs and LNNs cannot represent dissipation by construction, so the conservation test is restricted to the undamped regime. Contamination from pendulum equations in training corpora cannot be excluded. Finally, the numerical results cover one system and three LLMs; extension to other chaotic systems and model sets is future work.

8 Conclusion

On a controlled k -pendulum benchmark, held-out evaluation substantially changes the learned-dynamics ranking: off training trajectories, learned models increase in error and do not exceed simple baselines. Among evaluated black-box methods, kimi has the lowest reliability-adjusted error; known-equation integrators remain the numerical reference. Reliability-adjusted scoring and matched disclosure controls change the interpretation and find little evidence of in-context system identification. The main contribution is the protocol rather than the version-specific ranking.

Reproducibility. Code, prompts (Appendix A), trained weights, held-out specification, and per-cell checkpoints are released. Training uses seed 42; held-out evaluation uses seed 20260620 with

180 trajectories. LLMs (kimi-k2.6, grok-4-1-fast-reasoning, deepseek-v4-pro; Azure AI Foundry; June 2026) are sampled once at temperature 0. Because hosted deployments can change, reported LLM numbers are reproducible from released checkpoints, while API reruns are new evaluations. Invalid cells are assigned $\pi/2$. Local runs use a single 8 GB GPU.

Author Contributions

S.N.K. (lead author, corresponding) designed the benchmark, led the evaluation harness and the learned-dynamics, time-series, and analysis pipelines, ran the local out-of-sample and significance experiments, and wrote the manuscript. T.S. helped plan and build the benchmark harness. Author order reflects relative contribution. All authors reviewed and approved the final manuscript.

Non-author contributions. Shrithik Shahapure provided compute access for the Azure AI Foundry large language model evaluations and helped run those API evaluations.

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NeurIPS Paper Checklist

1. **Claims.** [Yes] The abstract, results, discussion, and limitations state the main claims and their scope.
2. **Limitations.** [Yes] Limitations are discussed in §7, including single held-out seed, model-scale asymmetry, Chronos history conditioning, learned-model training budget, and system scope.
3. **Theory assumptions and proofs.** [N/A] The paper is an empirical benchmark and does not present theoretical proofs.
4. **Experimental reproducibility.** [Yes] §4 and the Reproducibility paragraph specify model families, seeds, scoring rules, held-out trajectories, and API/version caveats.
5. **Open access to data and code.** [Yes] The repository includes code, prompts, held-out specifications, summaries, and trained local-model artifacts; large raw checkpoints may be distributed as release assets.
6. **Experimental setting details.** [Yes] The benchmark grid, regimes, inputs, metrics, and model classes are specified in §4.
7. **Statistical significance.** [Yes] Confidence intervals, paired tests, bootstrap intervals, and Holm correction are reported for the main comparisons.
8. **Compute resources.** [Yes] The paper reports local 8 GB GPU use and the estimated \$110.38 nominal API cost for the LLM evaluation.
9. **Code of ethics.** [Yes] The work is a scientific forecasting benchmark using simulated physical systems and does not involve human subjects.
10. **Broader impacts.** [N/A] The benchmark studies simulated dynamics and has no direct deployment setting.
11. **Safeguards for high-risk assets.** [N/A] The paper releases benchmark code/artifacts, not a high-risk generative model.
12. **Licenses for assets.** [Yes] The project includes an open-source license; external model APIs and cited libraries retain their own terms.
13. **New assets.** [Yes] The benchmark artifacts are generated by the authors from simulated dynamics, with generation code and seeds provided.
14. **Human subjects.** [N/A] No human subjects or private personal data are used.
15. **Institutional review board.** [N/A] No human-subjects research is performed.

A Prompt templates

The prompts (from `bench/prompts.py`) are reproduced below.

System prompt (direct / no-CoT).

You are a careful physicist predicting the future state of a k -pendulum (a chain of k point masses on massless rods, hinged in a planar chain, swinging under gravity with viscous damping). Respond with ONLY a single JSON object on one line, no prose, no code fences. Schema:
`{"theta": [...k floats..], "omega": [...k floats..]}`
 θ_i is in radians (link i measured from straight-down, positive counter-clockwise)
and ω_i is angular velocity in rad/s.

System prompt (chain-of-thought). Same preamble, then: “Reason step-by-step about the dynamics, then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a single JSON object: `{"theta": [...k floats..], "omega": [...k floats..]}`.”

User prompt (coordinate modality).

Number of links $k = \{k\}$
Physical constants:

Table 7: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.458	1.517

```

g (gravity, m/s^2): {g}
L (rod lengths, m): {L}
m (point masses, kg): {m}
damping (viscous coefficient on omega): {damping}
Initial state at t=0:
  theta_0 (rad): {theta0}
  omega_0 (rad/s): {omega0}
Predict the state at t = {T} seconds.
Return theta and omega as length- $\{k\}$  JSON arrays.

```

Hidden-constants regime. The constants block above is replaced by:

Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point masses, and damping are NOT disclosed. You must infer dynamics from the initial conditions alone.

Parsing and scoring. Each response is stripped of code fences and (for CoT) of everything outside the `<answer>...</answer>` tags; the first `{...}` object is JSON-parsed and validated to contain length- k theta and omega arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

B Training and compute

All learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, on a single NVIDIA GTX 1070 Ti (8 GB). Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at $\Delta t = 0.01$ s from the 10^{-3} s ground truth. These samples are drawn from *nine* trajectories—one per (k, regime) cell—so, while the sample count is large, the number of distinct trajectories the learned models see is small.

C Per- k and per-regime breakdowns

Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), by number of links (Table 7) and by regime (Table 8); models in leaderboard order. Cells a model never produced (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$, so each row is a mean over the same grid as Table 3.

Table 8: Out-of-sample error by regime. Learned-model hidden cells use the equalized no-hidden-constants run. The hidden regime remains easier for several models because its low-gravity/short-rod constants give slower, less-divergent motion; this is why the unadjusted regime contrast (§5.6) is a difficulty confound rather than evidence of in-context system identification.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.511	1.408	1.127

Table 9: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn (full-scale)	1.259	1.439

D Additional result tables

Full per-model values for the horizon breakdown (§5.2) and the predictability horizon (§5.6); the key numbers are stated inline in the body.

Table 10: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better.

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01